

Automated Docking of Articulated Vehicles: Advances in Control, Coordination, and Experimental Validation

1. Automated Docking of Articulated Vehicles in Confined Environments: An Overview

1.1. The Logistics Imperative and Automation Challenges

The global logistics industry is undergoing a significant transformation, driven by factors such as the growth of e-commerce and globalization, with the market projected to reach \$8.04 trillion by 2030.¹ Within this landscape, road transport, particularly using tractor-semitrailer combinations, forms the backbone of freight movement, especially in regions like the European Union where they account for approximately 77.8% of total tonne-kilometers performed by road freight vehicles.¹ However, the sector faces mounting pressures from demands for faster deliveries, rising operational costs, driver shortages, and the urgent need for environmental sustainability.¹ Automation in distribution centers (DCs) presents a compelling solution to many of these challenges, offering the potential for increased efficiency, reduced human error, and optimized operational workflows.¹

Automating articulated vehicles, specifically tractor-semitrailers, for docking maneuvers in the confined and dynamic environments of DCs is a particularly complex endeavor.¹ The inherent characteristics of these vehicles and their operational context give rise to several significant challenges:

- **Instability During Reverse Maneuvers:** Articulated vehicles are inherently unstable when reversing due to the articulation joint between the tractor and semi-trailer. This instability necessitates continuous steering corrections by the driver (or an autonomous system) to prevent jackknifing, where the trailer swings out and potentially collides with the tractor.¹ This phenomenon is a primary concern for low-speed reverse operations.¹
- **High Precision Requirements:** Docking operations demand a high degree of precision to align the semi-trailer accurately with the loading gate, often with tolerances in the order of centimeters for lateral position and sub-degree accuracy for orientation.¹
- **Confined Maneuvering Spaces:** DCs are typically characterized by narrow lanes, tight corners, and limited space for maneuvering large vehicles. This spatial constraint, coupled with the large swept path of articulated vehicles, significantly complicates path planning and execution.¹
- **Complex Low-Speed Dynamics:** Maneuvers in DCs involve both forward and

reverse driving at low speeds, often requiring large steering and articulation angles. Accurately predicting and controlling vehicle behavior under these conditions is non-trivial.¹

The confluence of these factors—inherent instability, stringent precision demands, and severe spatial constraints—defines a unique and formidable problem set for automation. Solutions must be specifically tailored to these unforgiving conditions, moving beyond generic autonomous driving algorithms to address the nuanced dynamics of articulated vehicles in close-quarters operations.

1.2. Research Pillars for Automated Docking Solutions

Addressing the multifaceted challenge of automated docking necessitates a comprehensive research approach, typically encompassing several interconnected pillars. As identified in the foundational work by Gosar (2025)¹, these pillars include vehicle modeling, path-following control, traffic management for multi-vehicle scenarios, and robust simulation and implementation strategies. Vehicle modeling provides the essential understanding of the system's behavior. Path-following control ensures the vehicle can accurately execute planned maneuvers. Traffic management extends this capability to coordinate multiple vehicles safely and efficiently. Finally, simulation and practical implementation, often involving scaled environments, validate the developed theories and algorithms in increasingly realistic settings.¹

While a thorough understanding of vehicle dynamics through modeling is a crucial starting point, the focus of this review, aligning with specific research directives, will be weighted towards the advancements in control methodologies, coordination strategies, and the practical realization and validation of these systems. The development and detailed evaluation of complex vehicle models, though foundational, will be summarized concisely to allow for a more in-depth exploration of how these models are leveraged for sophisticated control and system-level operation. This emphasis reflects a desire to understand the subsequent stages of autonomous system development: once a sufficiently representative model is established, how are robust control and coordination achieved and proven in practice?

2. Foundational Vehicle Modeling for Docking Applications (Concise Summary)

2.1. The Role of Vehicle Models in Automated Docking

The development of any autonomous vehicle system begins with the creation of mathematical models that describe its motion. These models are indispensable for

understanding vehicle behavior, designing effective control strategies, and creating simulation environments for testing and validation.¹ For the specific application of automated docking of articulated vehicles, models must capture the vehicle's response during low-speed forward and reverse maneuvers, often involving large steering and articulation angles—conditions prevalent in constrained DC environments.¹

The research by Gosar (2025) evaluated three distinct vehicle models: a kinematic model, a non-linear single-track dynamic model (STM), and a high-fidelity multibody dynamic model (MBD).¹ The kinematic model, being the simplest, primarily describes the geometric motion of the vehicle without explicitly considering forces or tire slip.¹ The STM incorporates vehicle mass properties and linear tire characteristics, offering a higher level of dynamic detail.¹ The MBD model, developed in tools like MATLAB Simscape, represents the highest fidelity, incorporating non-linear tire behavior and numerous degrees of freedom.¹

Evaluations against real-world data collected from low-speed maneuvers in a DC revealed trade-offs. While the MBD model demonstrated the highest accuracy in forward driving (4.83% error for yaw rate and lateral velocity) and the STM model proved most accurate for reverse driving (27.7% error with the least steering effort), a strategic decision was made in Gosar (2025) to utilize the **kinematic model** for the subsequent design of the path-following controller.¹ This choice, despite the kinematic model's acknowledged lower accuracy in open-loop predictions (e.g., 8.3% cumulative lateral position error in forward driving due to neglecting tire slip, and requiring 27% additional steering effort in reverse)¹, was driven by several practical considerations. The kinematic model offers simplicity and significantly lower computational load (90% faster than MBD).¹ Crucially, its structure facilitates the derivation of **analytical expressions for controller gains**.¹ This characteristic is paramount for developing adaptable and easily tunable control systems, a key objective for the automated docking application. This engineering trade-off underscores a shift in priority from achieving perfect open-loop model prediction to enabling effective and analytically tractable closed-loop control design. The focus becomes how well the controller can perform using the model, rather than how well the model predicts behavior in isolation.

2.2. Scaled Vehicle Platforms for Development and Validation

Bridging the often-significant gap between simulation-based research and deployment on expensive, potentially hazardous full-scale vehicles is a critical challenge in autonomous systems development. Scaled vehicle platforms offer a

pragmatic and effective solution, providing a safe, cost-efficient, and controllable environment for the iterative development, testing, and validation of automated driving functionalities, particularly for operations in confined spaces like DCs.¹

The TruckLab at Eindhoven University of Technology, a 1:13.3 scaled replica of a DC environment featuring scaled tractor-semitrailers, exemplifies such a platform.¹ The core purpose of these scaled environments is to ensure that algorithms and control strategies developed and refined at a smaller scale can be reliably transferred to their full-scale counterparts with minimal modifications.¹ This requires rigorous validation of the scaled vehicles.

In Gosar (2025), the TruckLab vehicles underwent a comprehensive evaluation process.¹ This involved applying principles of dimensional analysis and similitude, ensuring geometric similarity (proportional scaling of dimensions) and kinematic similarity (proportional scaling of velocity and acceleration) between the scaled models and full-scale vehicles.¹ The dynamic behavior of the scaled vehicles during low-speed forward and reverse maneuvers was compared against both real-world measurement data from full-scale tractor-trailers and the predictions of the kinematic vehicle model.¹ Adjustments, such as implementing a wheelbase compensation factor in steering inputs and modeling steering system asymmetries, were made to enhance the fidelity of the scaled representation.¹ The findings confirmed that the scaled vehicles could accurately replicate the essential driving dynamics of full-scale vehicles in DC-like scenarios and that controllers based on the kinematic model could be effectively tested and utilized in this environment.¹ The successful validation and utilization of such scaled testbeds represent a vital methodological advancement, accelerating the research and development cycle for complex robotic applications by enabling robust testing in a high-fidelity, yet manageable, setting.

3. Advanced Path-Following Control Systems for Articulated Vehicles

The core of automating docking maneuvers lies in the development of sophisticated path-following control systems capable of precisely guiding the articulated vehicle along a desired trajectory, often under challenging conditions.

3.1. Control Objectives and Performance Requirements in Docking

The primary control objective for automated docking is to achieve accurate path following, guiding the semi-trailer to the docking gate within the tight confines of a DC and typically within a limited travel distance.¹ The design of such controllers is heavily influenced by specific performance requirements, which are often derived

from analyzing human expert driver behavior and the operational necessities of the docking process.¹

Gosar (2025) outlines a stringent set of performance criteria for the final phase of docking, based on full-scale real-world considerations¹:

- **Lateral Trailer Position Error (e_{y2bd}):** Less than 10 cm from the desired path.
- **Trailer Orientation Error ($e_{\psi2bd}$):** Within 0.5 degrees of the desired orientation.
- **Articulation Angle Error (e_{ybd}):** Within 0.5 degrees of the desired articulation angle.

These accuracy targets must be met and maintained once the vehicle is within a specified traveled distance ($s_{t,max}$), typically around 40 meters from the dock.¹

Furthermore, the controller must operate while respecting critical vehicle constraints:

- **Maximum Articulation Angle (γ_{max}):** Not exceeding 80 degrees to prevent jackknifing.
- **Maximum Steering Angle (δ_{max}):** Limited to ± 40 degrees.
- **Maximum Steering Rate ($\dot{\delta}_{max}$):** Limited to 40 degrees per second.

A crucial aspect of the controller design is its ability to seamlessly manage both forward and reverse driving modes, as docking maneuvers often involve both.¹ The definition of these performance requirements, derived from real-world operational needs, establishes a high benchmark for the controller. A significant design choice highlighted in the research is the parameterization of controller states and inputs based on spatial coordinates (traveled distance) rather than time.¹ This decouples the lateral path-following task from the longitudinal velocity control, granting operational flexibility. For instance, the vehicle can vary its speed or even stop (e.g., for an unexpected obstacle) without deviating from its designated geometric path, a vital capability in dynamic DC environments.¹

3.2. Path-Following Controller Design and Gain Determination

Given the operational demands of a DC, where vehicles must vary speed and potentially stop without path deviation, a path-following control strategy is favored over trajectory tracking (which is time-dependent).¹ The controller detailed in Gosar (2025) employs a feedback-feedforward architecture.¹ The feedback component utilizes errors in the trailer's lateral position (e_{y2}), the trailer's yaw angle ($e_{\psi2}$), and the articulation angle (e_{y1}) relative to the reference path.¹ Notably, the control point for both forward and reverse driving is consistently chosen as the semi-trailer axle. This strategic choice avoids discontinuities or "jumps" in the control reference when transitioning between driving directions, thereby minimizing potential initial off-track

errors.¹

For the design of the feedback controller, the non-linear error dynamics (derived from the kinematic model) are linearized around the reference path, assuming small errors and small angle approximations. These linearized dynamics are then expressed as a function of traveled distance (s) rather than time, reinforcing the decoupling of lateral and longitudinal control.¹ The steering input δ is then computed based on these errors.

A key innovation in the controller design is the use of the **pole placement method** to derive analytical expressions for the feedback controller gains (K_y, K_ψ, K_γ).¹ These gains are explicitly functions of the vehicle's critical dimensions (tractor wheelbase L_1 , kingpin to tractor drive axle distance L_{1c} , and trailer wheelbase L_2), the sign of the tractor's longitudinal velocity ($\text{sign}(v_{x1})$ which differentiates between forward and reverse motion), and the desired closed-loop eigenvalues ($\lambda_1, \lambda_2, \lambda_3$) of the linearized system.¹ This analytical approach offers significant advantages:

- **Adaptability:** The controller can be readily adapted to different configurations of single articulated vehicles (e.g., passenger cars with trailers, different truck-trailer combinations) by simply adjusting the dimensional parameters in the gain equations.¹
- **Straightforward Tuning:** Gain determination is simplified, as it directly relates to desired closed-loop dynamic characteristics (poles) and known vehicle geometry.

This analytical gain structure is a direct benefit of the earlier decision to use the kinematic model for controller design. While a higher-fidelity dynamic model might offer more precise open-loop predictions, the kinematic model's relative simplicity enables this elegant and adaptable gain formulation. This reflects a design philosophy that prioritizes controller robustness and ease of application across various vehicle types over the complexities that might arise from control design based on more intricate models.

The controller also incorporates a feedforward component, δ_{ff} , which utilizes the curvature of the tractor's reference path, κ_{1r} , to anticipate necessary steering actions.¹ The selection of the closed-loop eigenvalues (λ_j) is critical; they must be chosen to ensure the system responds quickly enough to meet performance criteria within the limited travel distance, yet not so aggressively as to exceed actuator limits (steering angle and rate constraints) or induce oscillations. Typically, real and equal eigenvalues are chosen to avoid the dominance of any single pole, which could lead to input saturation and instability.¹

The specific performance requirements and constraints that the path-following controller must adhere to are summarized in Table 1.

Table 1: Path-Following Controller Performance Requirements & Constraints

Parameter	Symbol	Value	Source(s)
Max. Lateral Trailer Position Error	e_{y2bd}	0.1 m	1
Max. Trailer Orientation Error	$e_{\psi2bd}$	0.5 deg	1
Max. Articulation Angle Error	e_{ybd}	0.5 deg	1
Max. Traveled Distance for Convergence	$s_{t,max}$	40 m	1
Maximum Articulation Angle	γ_{max}	80 deg	1
Maximum Steering Angle	δ_{max}	40 deg	1
Maximum Steering Rate (Time Domain)	$\dot{\delta}_{max}$	40 deg/s	1
Maximum Steering Rate (Distance Domain)	δ_{max}'	40 deg/m (example value from TR study)	1

These quantified targets and operational boundaries are central to the controller's design and evaluation, defining clear metrics for successful docking.

4. Enhancing Robustness and Adaptability: Hybrid Control and Maneuver Assessment

While a well-tuned path-following controller can perform effectively under ideal conditions, real-world docking scenarios are often fraught with uncertainties, such as

initial positioning errors, sensor noise, or unmodeled disturbances.¹ These can compromise the controller's ability to meet stringent docking precision requirements in a single attempt.

4.1. Addressing Real-World Docking Uncertainties

Continuous reverse maneuvering under adverse conditions or with accumulating errors significantly increases the risk of jackknifing, an unstable and potentially dangerous state for articulated vehicles.¹ To address this, Gosar (2025) proposes a hybrid control approach that emulates a common strategy employed by human drivers: if a reverse maneuver becomes problematic, the driver will often drive forward briefly to realign the vehicle relative to the target before attempting the reverse motion again.¹ This explicit acknowledgment of potential single-attempt failures and the incorporation of a human-like corrective strategy represents a significant step towards more pragmatic and robust autonomous systems. Such systems are designed not just for ideal execution but also for intelligent recovery from non-ideal situations, a crucial capability for deployment in dynamic and unpredictable DC environments.

4.2. The Tracking Region (TR) Concept

Central to this hybrid control strategy is the concept of a **Tracking Region (TR)**.¹ The TR is defined as the set of initial vehicle error states—specifically, the lateral trailer position error (e_{y20}), trailer orientation error ($e_{\psi20}$), and articulation angle error ($e_{\gamma0}$)—from which the path-following controller is capable of successfully guiding the vehicle to meet the specified docking performance requirements (as detailed in Table 1) within the remaining distance to the docking gate (D2D). This must be achieved while adhering to all vehicle state and actuation constraints (e.g., maximum steering angle $\delta_{\max}$, maximum steering rate $\dot{\delta}_{\max}$ in the distance domain).¹

A key aspect of the TR developed in Gosar (2025) is its **analytical determination**.¹ By solving the linear Ordinary Differential Equations (ODEs) that describe the linearized path-error dynamics¹, the boundaries of this multi-dimensional TR can be computed in real-time.¹ This contrasts with computationally expensive numerical methods often used to estimate regions of attraction for non-linear systems.¹ The TR's boundaries are shaped by the performance error bounds (X_{bd}), the actuation limits ($\delta_{\max}, \dot{\delta}_{\max}$), the chosen closed-loop eigenvalues (λ_j), and the current D2D.¹ As the vehicle gets closer to the dock (D2D decreases), the TR naturally shrinks, reflecting the reduced margin for error and correction.¹

The TR essentially provides the controller with a form of "self-awareness"—a predictive capability to assess whether it can succeed from its current state given the

remaining maneuver distance. Instead of simply reacting to errors as they occur, the controller can proactively determine if the current situation is recoverable by the reverse path-following logic alone. This predictive assessment is fundamental to the intelligent decision-making process of the hybrid control system. The key parameters defining this operational envelope are summarized in Table 2.

Table 2: Key Parameters and Constraints for Tracking Region (TR) Definition

Parameter	Symbol	Typical Value/Range (Full-Scale)	Influence on TR	Source(s)
Lateral Position Error Bound	e_{y2bd}	0.1 m	Defines acceptable final lateral error	¹
Trailer Orientation Error Bound	$e_{\psi2bd}$	0.5 deg	Defines acceptable final orientation error	¹
Articulation Angle Error Bound	e_{ybd}	0.5 deg	Defines acceptable final articulation error	¹
Maximum Steering Angle	δ_{max}	40 deg	Constrains the maximum allowable steering input, shaping the TR boundary	¹
Maximum Steering Rate (Distance Dom.)	δ_{max}'	40 deg/m	Constrains the maximum rate of steering change, shaping the TR boundary	¹
Closed-Loop Eigenvalues	λ_j	e.g., 0.188 (real, equal)	Determine system response speed and stability;	¹

			significantly impact TR size	
Distance to Dock	D2D	Variable (e.g., 40 m down to 0 m)	TR size decreases as D2D decreases, reflecting reduced maneuverability space	¹

4.3. Hybrid Path-Following Control Strategy

The hybrid control strategy leverages the TR for intelligent switching between driving modes.¹ The logic is straightforward:

1. If the vehicle's current error state (relative to the reference path) falls **within** the dynamically calculated TR for the current D2D, the reverse path-following controller remains active, and the vehicle continues its reverse maneuver towards the dock.¹
2. If the vehicle's error state moves **outside** the TR, it signifies that the reverse controller is unlikely to meet the performance requirements from that state within the remaining distance without violating constraints. In this case, the system triggers a switch to a forward driving mode. The vehicle then follows a predefined forward path (e.g., a straight line perpendicular to the dock) to realign itself.¹

To prevent "chattering"—rapid, repeated switching between forward and reverse modes when the vehicle state hovers near the boundary of the TR, potentially due to sensor noise or small oscillations—a **conservative region (Sc)** is introduced.¹ This Sc is typically defined as a region slightly smaller (e.g., 30% smaller in error tolerances and actuation limits) than the main TR. When switching from forward back to reverse, the vehicle must first enter this more restrictive Sc before the reverse maneuver is re-initiated.¹ This hysteresis in the switching logic ensures smoother and more decisive transitions. Furthermore, during these mode switches, the steering commands are interpolated over a short transition distance to ensure smooth actuation and avoid abrupt changes.¹

This hybrid control framework, underpinned by the predictive capability of the TR, elevates the automated docking system beyond a simple path follower. It becomes a more intelligent agent capable of strategic maneuvering, self-assessment, and recovery from non-ideal conditions, thereby significantly enhancing its robustness and practical utility in the complex and often unpredictable environment of a

real-world DC.

5. Coordinating Multiple Vehicles: Path Planning and Traffic Management in Distribution Centers

The automation of individual vehicle docking is a significant step, but in a functional DC, multiple vehicles operate concurrently. This necessitates robust systems for path planning and traffic management to ensure safe, efficient, and deadlock-free operations.¹ Chapter 6 of Gosar (2025) addresses this by proposing an offline path planning algorithm and an online traffic management system tailored for the TruckLab DC layout.¹

5.1. Offline Path Planning for DC Operations

For the structured and repetitive nature of tasks within a DC (e.g., moving from a parking spot to a specific dock), an offline path planning approach can be highly effective. The objective detailed in Gosar (2025) is to generate a **static library of kinematically feasible reference paths** for all predefined routes within the TruckLab DC environment.¹ This library serves as a repository of pre-computed trajectories that vehicles can select from during online operations based on their assigned tasks. This approach reduces the online computational burden compared to dynamic path planning, which might be critical in a multi-vehicle system.

The **waypoint generation** process is tailored to the specific layout of the TruckLab DC and the nature of the tasks¹:

- **Inputs:** The algorithm takes the global DC map, along with specific parking entry/exit numbers (e.g., pen_i, pex_k) and docking gate numbers (e.g., doc_j) as inputs.
- **Task Phases:** For a task like moving from a parking entry to a docking gate (Task1), waypoints are generated in distinct phases:
 1. **Forward Driving:** Waypoints guide the vehicle from the parking entry towards an approach area near the docking gates. Heuristics are used for placement, such as regular spacing along centerlines at specific angles (e.g., 45° or 90°), and offsets are introduced at intersections to avoid sharp turns and ensure sufficient maneuvering space for the trailer.¹ Waypoints might extend beyond the target dock to facilitate better alignment for the subsequent reverse maneuver.
 2. **Reverse Driving:** For the docking phase, forward waypoints are often reused in reverse order. These reverse waypoints are further refined through piecewise interpolation to include the trailer's desired yaw angle (ψ_{2r}),

providing more comprehensive guidance for the unstable reverse motion.¹

3. **Complementary Maneuvers:** To support the hybrid control strategy (Chapter 5), complementary straight-line waypoints are generated perpendicular to the docking gate (both forward away from the gate and reverse back to it). These provide an alternative path for realignment if the initial docking attempt is compromised.¹
- **Reference State Generation:** To transform these coarse waypoints into the detailed reference states required by the path-following controller (i.e., trailer pose ($X_{2r}, Y_{2r}, \psi_{2r}$), articulation angle γ_r , and tractor reference curvature κ_{1r}), simpler lateral control algorithms are used to guide a kinematic vehicle model along the waypoints.¹ For forward segments, a **Pure Pursuit** strategy is employed with the tractor's rear axle as the tracking point. For reverse segments, an **Inverse Kinematics** approach is used, with the semi-trailer axle as the tracking point and a virtual steering angle guiding the trailer.¹

It is important to note that these lateral controllers (Pure Pursuit, Inverse Kinematics) are part of the *offline path generation process* for the reference library. They are distinct from the more sophisticated path-following controller¹ that the actual vehicle uses for real-time tracking. This offline generation of a static path library is a pragmatic choice for the DC environment. While some literature explores complex online path planning algorithms like optimization-based methods⁹, Rapidly-exploring Random Trees (RRT/RRT*)¹⁶, or lattice-based planners with motion primitives¹⁹, the fixed nature of routes in many DC operations makes a pre-computed library a viable and computationally efficient alternative. This approach prioritizes predictability and reduces the real-time processing load on the individual vehicles within a multi-agent system.

5.2. Online Traffic Management for Multiple Articulated Vehicles

With a library of reference paths available, the next challenge is to coordinate multiple vehicles operating simultaneously. Gosar (2025) proposes a **hybrid control architecture** for this purpose, where decision-making responsibilities are shared between a central system and the individual vehicles.¹ This architecture integrates dispatching, scheduling, path selection, and traffic management components.

The core of the traffic management strategy is **zone-based control**.¹ The DC map (TruckLab layout) is divided into a set of discrete zones (e.g., 14 zones, z_1 to z_{14}), where each zone can typically accommodate one vehicle at a time. The minimum zone length is designed to be approximately the wheelbase of a tractor-semitrailer. To account for the full vehicle length (including overhangs), an additional safety measure

involves reserving the zone previously occupied by a vehicle.¹ The system employs multiple levels of zones:

- **Standard Zones:** Most zones (z_i) allow only one vehicle.
- **Multi-Vehicle Zones:** Certain zones, like parking areas (z_1, z_{12}), are designated to accommodate multiple stationary vehicles simultaneously.¹
- **Safety Zones (sj):** Associated with each docking gate (doc_j), these are crucial for preventing deadlocks and collisions, especially when one vehicle is maneuvering in the docking approach area (zones z_5 to z_8) while another might be exiting an adjacent dock.¹

Vehicles share their **zone status** (e.g., which zones are occupied, reserved by them, or targeted by them) with each other to enable coordination.¹ While this global sharing works for the TruckLab scale, for larger DCs, sharing information about only nearby zones might be more computationally efficient.

The roles within this hybrid architecture are distributed:

- **Central System (Dispatcher):**
 - Manages overall operations by assigning tasks to individual vehicles based on a predefined Job Sheet (defining sequences like $peni \rightarrow doc_j \rightarrow pex_k \rightarrow peni$) and the global Zone Layout.¹
 - Monitors vehicle pose and task completion status to make informed task allocations. For example, if a vehicle completes a task and arrives at a designated hand-off zone (e.g., a dock, a parking exit), the dispatcher assigns the next task from its job sheet.¹
- **In-Vehicle Traffic Management:**
 - **Path Selection:** Based on the task allotted by the Dispatcher, the vehicle selects the appropriate predefined reference path from its local Static Reference Path Library.¹
 - **Scheduler:** Sets the vehicle's operating mode (forward, reverse, stop) based on task progress and incorporates fixed idle times for simulated loading/unloading, parking, or transitioning between forward and reverse maneuvers (mimicking human driver behavior).¹
 - **Zone Control:** Implements the rules for zone occupancy and progression. This typically involves a "one-zone look-ahead" policy: before entering a new target zone, the vehicle checks if it's free. If occupied, the vehicle waits. Specific rules govern forward and reverse movements¹:
 - *Forward Movement:* Current zone marked occupied, previous zone reserved for safety.
 - *Reverse Movement:* Target zone in reverse is reserved. Reverse operations

are typically restricted to specific zones (e.g., z5–z8 for docking approach).

- *Priority*: Vehicles performing reverse maneuvers (e.g., docking) are often given priority over those moving forward in shared conflict areas.¹
- *Dock Exiting*: Before a vehicle exits a dock docj to start its next task, it checks if adjacent safety zones (sj-1,sj,sj+1) are clear to avoid conflict with vehicles maneuvering into neighboring docks.¹

The path-following controller¹ is then utilized by the vehicle to track the selected reference path segment, using the drive mode (forward/reverse) and target velocity provided by the local Traffic Management system.¹ This hybrid architecture, combining centralized task assignment with decentralized, zone-based conflict resolution, aims to provide a balance between global coordination and local autonomy, a common and often effective pattern in managing multi-robot or multi-AGV systems in structured environments like warehouses or terminals.¹

6. From Simulation to Reality: Experimental Validation and Implementation

A cornerstone of robust robotics research is the progression from theoretical development and simulation to practical implementation and validation in environments that increasingly approximate real-world conditions. Gosar (2025) follows this paradigm, extensively using both simulation and the TruckLab scaled experimental platform.

6.1. Validation of Scaled Testbeds

As previously discussed (Section 2.2), scaled testbeds like TruckLab play a pivotal role in the development and validation of autonomous driving functionalities, especially for complex maneuvers in constrained environments where full-scale testing is initially prohibitive due to cost, safety concerns, and logistical complexity.¹ The credibility of results obtained from such platforms hinges on the rigorous validation of their ability to accurately represent full-scale vehicle behavior.¹

The validation process for the TruckLab vehicles, as detailed in Gosar (2025)¹, involved several key steps:

- **Dimensional Analysis and Similitude**: Principles of geometric similarity (scaling of physical dimensions by a factor of 1:13.3) and kinematic similarity (corresponding scaling of velocities and accelerations) were applied to ensure that the motions of the scaled vehicles correspond to their full-scale

counterparts.¹ Mass was scaled by L3, assuming similar material density, and time and angles (and their rates) remained unscaled, supporting the goal of controller transferability.¹

- **Comparative Analysis:** The behavior of the scaled tractor-semitrailers was meticulously compared against both full-scale real-world measurement data (collected from a DAF XF450 tractor and a standard semi-trailer during low-speed forward and reverse maneuvers at a DPD distribution center) and the predictions of the kinematic vehicle model.¹
- **Characteristic Matching and Adjustments:** To enhance representational accuracy, specific characteristics observed in full-scale vehicles were incorporated into the scaled vehicle setup and the kinematic model used for comparison. This included modeling the asymmetric steering response (due to tie-rod geometry) using a quadratic function of the steering wheel angle and accounting for steering offsets observed at zero path curvature.¹ For comparing paths where tractor wheelbases differed slightly between the scaled-up lab vehicle and the real test vehicle, a wheelbase compensation factor was applied to the steering input of the scaled vehicle to align yaw rates.¹ Furthermore, to ensure fair comparison during unstable reverse maneuvers, a proportional feedback controller was implemented to stabilize the articulation angle in both the scaled vehicle and the kinematic model when replaying real-world data.¹

The results of this validation process confirmed that the scaled lab vehicles, when operated under these similarity principles and with appropriate characteristic adjustments, could indeed replay full-scale measurement data with acceptable fidelity for low-speed maneuvers.¹ It also supported the conclusion that kinematic model-based controllers, designed and tested in the TruckLab environment, could be effectively utilized and their findings extrapolated to full-scale applications.¹ This rigorous validation is crucial, as it underpins the reliability of all subsequent experimental work conducted on the scaled platform, ensuring that the insights gained are not mere artifacts of the scaled environment but are genuinely indicative of full-scale system behavior.

6.2. Experimental Testing of Control and Coordination Algorithms

Following the validation of the TruckLab platform, Gosar (2025) systematically tested the developed control and coordination algorithms, progressing from simulation to scaled hardware experiments for each major component of the automated docking system.¹

- **Path-Following Controller**¹: The performance-aware path-following controller was first evaluated in a simulation environment using the non-linear kinematic

model. Subsequently, it was implemented and tested on the scaled vehicles in TruckLab. These experiments demonstrated the controller's ability to accurately track straight-line and more complex docking paths (involving both forward and reverse segments), meeting the stringent performance requirements for lateral position and orientation errors while respecting actuation limits.¹ Video evidence showcased the smooth operation of the scaled vehicle under this controller.¹

- **Hybrid Control with Tracking Region**¹: The hybrid control strategy, incorporating the Tracking Region (TR) concept, was also validated through both simulation and TruckLab experiments. Tests were designed to evaluate the system's response to initial positioning errors that placed the vehicle outside the TR. The results confirmed that the system could correctly identify such situations, trigger a switch to forward motion for realignment, and then successfully re-engage the reverse path-following controller to complete the docking maneuver once the vehicle was back within the (conservative) TR.¹ These experiments highlighted the practical effectiveness of the TR in enhancing robustness.
- **Multi-Vehicle DC Management System**¹: The offline path planning and online traffic management algorithms for coordinating multiple vehicles were tested in simulation and on the three scaled vehicles in TruckLab. Various scenarios were explored, including:
 - All vehicles moving at the same constant velocity.
 - One vehicle experiencing a disturbance during docking, triggering a complementary realignment maneuver (leveraging the hybrid control logic) and demonstrating the system's ability to manage the ripple effects on other waiting vehicles.
 - Vehicles operating at different forward speeds, showcasing the traffic management's capability to handle non-uniform vehicle behavior.¹ The Gantt charts comparing zone usage over time between simulation and lab experiments showed consistent logical behavior, although some timing discrepancies due to sensor processing and communication delays in the lab setup were noted.¹ The experiments successfully demonstrated the zone-based conflict resolution and task progression according to the defined rules.¹

Throughout these experimental phases, the scaled TruckLab environment proved invaluable. It not only allowed for the verification of the algorithms' intended logic but also helped identify real-world implementation challenges, such as the impact of sensor noise on articulation angle control accuracy¹ and the effect of communication and processing latencies in multi-vehicle coordination.¹ This iterative loop of

simulation, scaled experimentation, and refinement is a hallmark of robust robotics research, ensuring that theoretical developments are grounded in practical applicability.

7. Synthesis and Future Outlook in Automated Articulated Vehicle Docking

7.1. Summary of Key Advancements and Contributions

The research synthesized in Gosar (2025) presents a comprehensive investigation into the automated docking of articulated vehicles, particularly focusing on the challenges encountered in constrained distribution center environments.¹ The work makes several key advancements and contributions to the field:

1. **Performance-Aware Path-Following Control:** A significant contribution is the development of a path-following controller specifically designed to meet stringent, real-world docking accuracy requirements (e.g., <10 cm lateral error, <0.5° orientation error within 40 m).¹ The controller's gains are derived analytically using pole placement, making them dependent on vehicle dimensions and desired closed-loop eigenvalues. This approach allows the controller to be adaptable for both forward and reverse driving and applicable to various single articulated vehicle configurations with straightforward tuning.¹ The decoupling of lateral and longitudinal control by operating in the traveled distance domain further enhances its practical utility.¹
2. **Robust Hybrid Control with Tracking Region (TR):** Addressing the inherent uncertainties and potential for single-attempt failure in reverse docking, the thesis introduces a novel hybrid control scheme.¹ This scheme incorporates the "Tracking Region" concept—an analytically determined, real-time assessable region of feasible operation for the reverse controller. If the vehicle state falls outside the TR, a human-like strategy of switching to forward motion for realignment is invoked, significantly enhancing the robustness and success rate of docking maneuvers under non-ideal conditions.¹
3. **Multi-Vehicle Coordination Framework:** The research extends beyond single-vehicle control to address multi-vehicle operations within a DC. An offline path-planning algorithm is developed to generate a static library of feasible reference paths for common DC tasks. This is coupled with an online traffic management system employing a hybrid architecture (centralized dispatch, decentralized zone-based conflict resolution) to coordinate multiple automated tractor-trailers.¹ This provides a foundational system for managing complex interactions in a shared workspace.

4. **Rigorous Experimental Validation via Scaled Platform:** A crucial aspect of the research is the extensive use and validation of the TruckLab scaled experimental platform.¹ By ensuring that the scaled vehicles accurately replicate full-scale low-speed dynamics, the thesis provides a credible pathway for testing and refining complex algorithms in a safe, cost-effective, and controlled environment before potential full-scale deployment. This systematic progression from simulation to scaled hardware experiments for each developed component (path-following, hybrid control, multi-vehicle management) underscores a practical and robust research methodology.¹

While detailed vehicle modeling was intentionally de-emphasized in this review as per the user's request, its foundational role in enabling the aforementioned control designs—particularly the choice of a kinematic model to facilitate analytical gain derivation—is an important underlying element of the thesis's practical approach.¹ Overall, the work provides a holistic framework, from low-level control design to multi-vehicle system coordination, tailored specifically to the demanding problem of automated docking for articulated vehicles. The emphasis on quantifiable performance, robustness through adaptive strategies, and thorough practical validation are its defining strengths.

7.2. Remaining Challenges and Future Research Directions

Despite the significant advancements presented, the field of automated articulated vehicle docking continues to offer numerous avenues for future research and development. The conclusions and recommendations from Gosar (2025)¹, along with trends identified in the broader literature, highlight several key areas:

Vehicle Modeling and Evaluation:

While the kinematic model proved useful for controller design, its limitations in accurately capturing all dynamic effects (like tire slip) were acknowledged.¹ Future work could involve:

- **Enhanced Sensor Suites and Testing:** Conducting a broader range of full-scale tests with higher-quality sensors to improve the accuracy of measured articulation angles and semi-trailer poses.¹
- **Observer Development:** Designing observers to robustly estimate critical states like articulation angle and trailer pose, potentially fusing data from multiple sensors to overcome individual sensor limitations.¹
- **Refined Evaluation Metrics:** For reverse driving, exploring alternative metrics or evaluation methods beyond stabilizing an unstable open-loop model with an additional controller could provide deeper insights into model fidelity.¹

Control Synthesis:

The developed controllers show promise, but further refinements can enhance their real-world applicability:

- **Impact of Real-World Imperfections:** Investigating the effects of sensor noise, varying sampling rates, and communication delays on controller performance before full-scale implementation.¹
- **Optimizing Hybrid Switching:** Developing metrics to determine the optimal forward driving distance required during a hybrid switch to guarantee docking success within a minimal number of switches, thus avoiding frequent direction changes.¹
- **Expanding Tracking Region Applicability:** Extending the TR concept to more complex curved paths, which could enable a wider range of maneuvers including forward parking and more intricate reverse docking scenarios.¹
- **Multi-Articulation Vehicles:** Applying and adapting control strategies for vehicle combinations with multiple articulation points (e.g., B-doubles), which are common in some logistics operations.¹

Perception and Localization:

Reliable perception and localization are paramount for safe and efficient automation:

- **Optimal Sensor Configuration:** Determining the most effective types and placements of sensors (on-vehicle and potentially infrastructure-based, like LiDAR at docking gates) for robust trailer pose estimation, especially given the challenges of tractor-trailer interoperability.¹
- **Environmental Modeling and Obstacle Avoidance:** Integrating advanced perception systems for dynamic environment mapping and robust detection and avoidance of both static (parked trailers, infrastructure) and dynamic obstacles (other vehicles, personnel).¹ Current environmental representations range from grid maps to more sophisticated geometric and semantic maps.³³

Motion Planning:

The thesis utilized a static path library, which is effective for structured environments but has limitations:

- **Dynamic Path Planning:** Incorporating dynamic path planning capabilities to allow vehicles to react to unforeseen obstacles or changes in the DC environment in real-time, rather than relying solely on pre-computed paths.¹ This could involve algorithms like A*, RRT*, lattice-based methods, or optimization-based approaches discussed in the literature.⁵ Techniques for swept path area minimization would also be relevant here.⁴⁶

Fleet Management and DC Operations:

Optimizing the overall DC workflow involves more than individual vehicle automation:

- **Advanced Scheduling Algorithms:** Developing more sophisticated DC scheduling strategies to optimize vehicle utilization, minimize waiting times, and maximize throughput, considering various operational scenarios and constraints.¹
- **Autonomous (De)Coupling:** Investigating technologies and strategies for autonomous coupling and decoupling of trailers, as many DCs utilize shunt tractors that require such operations.¹
- **Standardization:** Promoting standardized software interfaces and communication protocols (V2V and V2I) to ensure interoperability between vehicles and infrastructure from different manufacturers.¹
- **DC Design for Automation:** Adapting DC layouts and traffic rules to better support automated vehicle operations, including dedicated corridors and mixed-traffic management strategies.¹

Emerging Trends and Broader Research Directions:

The field is rapidly evolving, with several emerging trends poised to significantly impact future developments:

- **Advanced AI/Machine Learning:**
 - **Learning-Based Path Planning and Control:** Exploring Deep Reinforcement Learning (DRL)¹², Imitation Learning (IL)⁵⁴, and hybrid AI approaches⁵³ for generating complex forward and reverse maneuvers, potentially leading to more adaptive and robust solutions than purely model-based methods. These techniques could learn from expert human driver data or through self-exploration in simulation.
 - **End-to-End Learning:** Investigating end-to-end learning frameworks that map sensor inputs directly to control actions, potentially simplifying the traditional modular pipeline of perception-planning-control, though challenges in interpretability and safety verification remain.⁵¹
- **Robust Planning Under Uncertainty:** Developing planners that explicitly account for various sources of uncertainty, such as sensor noise, unpredictable movements of other agents, and inaccuracies in environmental models, to generate guaranteed safe and efficient paths.³⁰
- **Human-Robot Interaction (HRI):** In DCs, autonomous vehicles will inevitably interact with human personnel (e.g., yard workers, other truck drivers). Future path planning and control systems must incorporate HRI considerations to ensure safe, predictable, and efficient collaboration in these mixed environments.⁶² This includes legible motion and clear communication of intent.
- **Improved Simulation Tools and Benchmarking:** The development of more sophisticated simulation tools⁶⁴ and standardized benchmarking datasets and scenarios specifically for low-speed maneuvering and docking of articulated

vehicles in DCs is crucial for comparative evaluation and progress in the field.⁶

The future of automated docking for articulated vehicles likely lies in a synergistic fusion of the robust model-based control techniques, such as those developed by Gosar (2025), with the adaptive and learning capabilities of advanced AI/ML methods. Combined with enhanced perception, dynamic planning, and sophisticated fleet management systems, these advancements will be key to realizing the full potential of automation in complex and demanding distribution center operations.

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